Relative Cowedges and Coends over a Site

Relative (co-)Yoneda Lemma for Categories Indexed over a Small-Generated Site

Indexed categories, fibrations, stacks and relative category theory

Indexed categories for relative topos theory

Yiqi

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Abstract

This phases 0–3 preview develops localization-sensitive cowedges and coends over a fixed site. Phases 2–3 prove an explicit fixed-base coequalizer formula, stable base change under chosen pullbacks and Beck–Chevalley, and a Fubini theorem under stable-inner existence. A counterexample on a pullback-deficient site records the exact boundary of the stability theorem. Relative co-Yoneda, profunctor composition, and site-presentation invariance are not claimed.

Contents

1 Introduction

Ordinary coends package dinatural families into a single universal object. They support the calculus of Kan extensions, profunctors, density, and the co-Yoneda lemma. For a category varying over a base, however, there are two different operations that one might call a coend. One can form an ordinary coend separately in each fibre, or one can ask for a universal object that is compatible with every change of base. The first operation is familiar but does not, by itself, produce the second. The elementary example in example 5.3 makes this distinction explicit.

This paper begins a coend calculus adapted to categories indexed over a Grothendieck site. Its basic move is to formulate the universal property over the whole slice site $\mathcal{C}/U$. A relative cowedge with apex X therefore has a component not only for every object in $\mathbb{D}(U)$, but for every generalized object $A \in \mathbb{D}(V)$ above every arrow $v : V \to U$. A second axiom compares these components after further pullback. The resulting universal object records precisely the descent-sensitive data that a fibrewise definition omits.

The motivation comes from Drieux's lecture series on indexed categories and relative topos theory [Dri25b, Dri25a, Dri26a, Dri26b]. Those slides give an indexed Yoneda construction, formulate indexed colimits by localization to every slice, and announce relative (co-)Yoneda and indexed-(co)limit projects. They are programme-level sources, not published proofs of the claims developed here. We therefore reconstruct the definitions from their universal properties and prove independently every existence or computation statement used below.

There is important prior art. Shulman's indexed-enriched formalism already constructs a canceling tensor product and the composition of indexed profunctors by a fibrewise coequalizer under indexed coproduct and Beck–Chevalley hypotheses [Shu13, Definition 2.4, Definition 2.10, and Lemma 3.25]. Earlier, Betti and Walters developed an end calculus for categories enriched in the bicategory of spans of a base topos [BW89]. Lee develops indexed profunctors over 2-categories [Lee23]. The present definition is narrower in one direction and different in another: it is initially unenriched and tied to a fixed site, but it makes localization over every slice part of the cowedge itself. Thus the intended new problem is not merely to decorate the ordinary coend sign. It is to relate coend calculus to topology, descent, and change of site in relative topos theory.

Results of phases 0–3

After fixing size and variance conventions, we:

1. define localized relative cowedges, coends, wedges, and ends;
2. identify a relative coend with a representing object for the cowedge functor;

3. construct restriction along every map of the site and the canonical base-change comparison between local coends;
4. prove invariance under a chosen indexed equivalence and reduction to the ordinary definition over the terminal site; and
5. exhibit a two-object indexed category for which fibrewise coends fail to assemble, while the localized coends are stable;
6. construct every fixed-base relative coend by an explicit coequalizer, using only the displayed coproducts, coequalizers, and left adjoints to reindexing;
7. prove stable base change when the base has chosen pullbacks, Beck–Chevalley holds, and reindexing preserves those colimits; and
8. prove a Fubini theorem whenever the inner relative coends exist and are stable under base change.

All these statements are proved in this paper. We do *not* yet claim a relative co-Yoneda lemma, a localization-sensitive profunctor bicategory, or invariance under a change of site presentation. Those are separated into explicit problems in section 7.

2 The ambient indexed data

2.1 Sites and relative-topos conventions

Fix a Grothendieck universe and a small site $(\mathcal{C}, J)$. Write

$$\mathcal{E} = \mathbf{Sh}(\mathcal{C}, J), \qquad \ell_J = a_J \mathsf{y} : \mathcal{C} \longrightarrow \mathcal{E}$$

for the associated sheaf topos and the sheafified Yoneda functor. The canonical stack attached to this presentation is the pseudofunctor

$$\mathbb{S}_{(\mathcal{C},J)} : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{CAT}, \qquad U \longmapsto \mathcal{E}/\ell_J(U),$$

whose reindexing functors are pullbacks. Caramello and Zanfa prove that this is a J-stack [CZ21, Definition 2.6.4 and Theorem 2.6.8]. It will be the principal coefficient system for applications, although the definition below works for a general indexed category of coefficients.

In the terminology of relative topos theory, a relative topos over $\mathcal{E}$ is a geometric morphism $p : \mathcal{F} \to \mathcal{E}$ [CZ21, Definition 8.2.2]. We keep this distinct from an indexed category $\mathbb{D}$: the latter is input from which relative categorical constructions are formed, not by itself a relative topos.

The smallness assumption is a phase-1 convention. A small-generated site is locally small and has a small J-dense subcategory [CZ21, p. 4]. After choosing such a subcategory one may apply the same formulas to the resulting small presentation; independence of that choice is a separate comparison theorem and is not assumed below. This distinction is essential: a formula on a presentation is not automatically an invariant of the base topos.

Remark 2.1 (Where the topology enters). The equations in definition 3.1 use the underlying category $\mathcal{C}$, not the covering families of J directly. The topology enters when $\mathbb{D}$ and $\mathbb{V}$ carry J-descent, when H respects that descent, or through the choice $\mathbb{V} = \mathbb{S}_{(\mathcal{C},J)}$. Without such hypotheses the construction is localization-sensitive but not yet topology-invariant.

2.2 Indexed categories, opposites, and coefficients

Let

$$\mathbb{D} : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Cat}, \qquad \mathbb{V} : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{CAT}$$

be pseudofunctors. We assume that each $\mathbb{D}(U)$ is small and that each $\mathbb{V}(U)$ is locally small. For an arrow $u : U' \to U$, both reindexing functors are denoted u^*; the ambient pseudofunctor will always be clear. The coherence isomorphism is written

$$\alpha_{v,u,X} : v^* u^* X \xrightarrow{\cong} (uv)^* X.$$

We suppress unit constraints when no ambiguity can result.

Let $\mathbb{D}^\vee$ denote the fibrewise opposite indexed category: its fibre at U is $\mathbb{D}(U)^{\mathrm{op}}$, with the induced pseudofunctorial coherence. This is the indexed opposite used in relative category theory [CZ21, Definition 2.1.4]. Fix an indexed functor

$$H : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \longrightarrow \mathbb{V}.$$

Thus each fibre has a functor

$$H_U : \mathbb{D}(U)^{\mathrm{op}} \times \mathbb{D}(U) \longrightarrow \mathbb{V}(U),$$

and every $u : U' \to U$ has an invertible comparison

$$\chi^H_{u,A,B} : u^* H_U(A,B) \xrightarrow{\cong} H_{U'}(u^*A, u^*B), \tag{1}$$

natural in A, B and coherent under composition. Requiring χ^H to be invertible is the point at which H is genuinely indexed rather than merely laxly indexed.

For $U \in \mathcal{C}$, the localization of $\mathbb{D}$ at U is

$$\mathbb{D}/U : (\mathcal{C}/U)^{\mathrm{op}} \longrightarrow \mathbf{Cat}, \qquad (v : V \to U) \longmapsto \mathbb{D}(V).$$

If $X \in \mathbb{V}(U)$, its cartesian constant section over $\mathcal{C}/U$ has value v^*X at $v : V \to U$. These two constructions supply the domain and apex of the relative cowedges below.

3 Localized cowedges and relative coends

Recall that for an ordinary functor $K : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathcal{B}$, a cowedge to X is a dinatural family $K(A,A) \to X$; its initial cowedge is the coend [Lor23, Definition 1.1.4 and Definition 1.1.6]. We add a base coherence condition to this familiar definition.

Definition 3.1 (Localized relative cowedge)**.** Fix $U \in \mathcal{C}$ and $X \in \mathbb{V}(U)$. A *relative cowedge from H to X over U* is a family

$$\omega_{v,A} : H_V(A,A) \longrightarrow v^*X \tag{2}$$

for every arrow $v : V \to U$ and every $A \in \mathbb{D}(V)$, satisfying:

(C1) *fibrewise dinaturality*: for each $f : A \to B$ in $\mathbb{D}(V)$,

$$\omega_{v,A}\, H_V(f^{\mathrm{op}}, 1_A) = \omega_{v,B}\, H_V(1_B, f) : H_V(B,A) \longrightarrow v^*X; \tag{3}$$

(C2) *base coherence*: for every $w : W \to V$,

$$\omega_{vw,w^*A}\, \chi^H_{w,A,A} = \alpha_{w,v,X}\, w^*(\omega_{v,A}) : w^* H_V(A,A) \longrightarrow (vw)^*X. \tag{4}$$

The set of such families is denoted $\mathrm{Cow}^{\mathrm{rel}}_U(H;X)$.

The definition ranges over the Grothendieck construction of $\mathbb{D}/U$, but writing out equations (3) and (4) avoids choosing a cleavage for that construction. The two equations have different jobs: dinaturality moves within one fibre, whereas base coherence moves between fibres.

A morphism $(X,\omega) \to (Y,\tau)$ is a morphism $m : X \to Y$ in $\mathbb{V}(U)$ such that

$$v^* m\, \omega_{v,A} = \tau_{v,A}$$

for all v, A. This defines the category $\mathbf{Cow}_U(H)$. Equivalently, $\mathbf{Cow}_U(H)$ is the category of elements of the covariant functor

$$\mathrm{Cow}^{\mathrm{rel}}_U(H;-) : \mathbb{V}(U) \longrightarrow \mathbf{Set}.$$

Definition 3.2 (Relative coend)**.** A *relative coend of H over U* is an initial object (L_U, λ^U) of $\mathbf{Cow}_U(H)$. When it exists, we write

$$L_U = \int^{A\in\mathbb{D}}_{\mathrm{rel},U} H(A,A).$$

The notation records that the cowedge is localized over $\mathcal{C}/U$; it is not an assertion that an ordinary fibrewise coend has already been computed.

Proposition 3.3 (Representing property and uniqueness)**.** *For $L_U \in \mathbb{V}(U)$, the following are equivalent.*

1. *There is a relative cowedge λ^U making (L_U, λ^U) initial in $\mathbf{Cow}_U(H)$.*
2. *There is a natural isomorphism*

$$\mathbb{V}(U)(L_U, X) \cong \mathrm{Cow}^{\mathrm{rel}}_U(H;X) \tag{5}$$

sending 1_{L_U} to λ^U.

Consequently, a relative coend is unique up to a unique isomorphism compatible with its universal cowedge.

Proof. Given an initial cowedge, send $m : L_U \to X$ to the family $(v^* m\, \lambda^U_{v,A})_{v,A}$. Initiality gives the inverse and its uniqueness, while composition in $\mathbb{V}(U)$ gives naturality in X. Conversely, the image of 1_{L_U} under equation (5) is a universal element, hence an initial cowedge. The final assertion is the standard uniqueness of representing objects, including compatibility with the distinguished universal element. □

Definition 3.4 (Relative wedge and end)**.** For an indexed functor $K : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \to \mathbb{V}$, a relative wedge from $X \in \mathbb{V}(U)$ to K consists of maps

$$v^* X \longrightarrow K_V(A,A)$$

that satisfy the equations dual to equations (3) and (4). A *relative end* is a terminal relative wedge. This draft takes the dual universal property as the definition; no completeness of the fibrewise opposite coefficient system is assumed.

4 Restriction and stable base change

The use of all generalized objects has an immediate payoff: cowedges restrict without any existence assumption.

Proposition 4.1 (Restriction to a slice)**.** *Every arrow $u : U' \to U$ induces a functor*

$$\rho_u : \mathbf{Cow}_U(H) \longrightarrow \mathbf{Cow}_{U'}(H).$$

It sends an apex X to u^*X. Up to the pseudofunctorial constraints, the component of $\rho_u\omega$ at $v : V \to U'$ and $A \in \mathbb{D}(V)$ is $\omega_{uv,A}$. These functors are pseudofunctorial: $\rho_{1_U} \cong 1$ and $\rho_v\rho_u \cong \rho_{uv}$.

Proof. Define

$$(\rho_u\omega)_{v,A} : H_V(A, A) \xrightarrow{\omega_{uv,A}} (uv)^*X \xrightarrow{\alpha^{-1}_{v,u,X}} v^*u^*X.$$

The dinaturality equation is exactly that for $\omega_{uv,-}$. Base coherence follows by pasting the base-coherence square for ω with the associativity coherence for α. On morphisms put $\rho_u(m) = u^*m$. The identity and composition comparisons are inherited from those of $\mathbb{V}$, and their coherence is inherited from the pentagon and triangle axioms of the pseudofunctor. □

Suppose relative coends (L_U, λ^U) have been chosen for all $U \in \mathcal{C}$. The restricted cowedge $\rho_u\lambda^U$ has apex u^*L_U. Initiality at U' therefore supplies a unique comparison

$$b_u : L_{U'} \longrightarrow u^*L_U \tag{6}$$

such that $b_u\lambda^{U'} = \rho_u\lambda^U$, with the structural isomorphisms understood.

Definition 4.2 (Stable relative coends)**.** A chosen family of relative coends is *stable under base change* if every canonical map b_u in equation (6) is an isomorphism.

Theorem 4.3 (Coherent assembly)**.** *The canonical maps b_u satisfy the identity and composition laws up to the coherence of $\mathbb{V}$. If the relative coends are stable, the inverses*

$$b_u^{-1} : u^*L_U \xrightarrow{\cong} L_{U'}$$

make $(L_U)_{U\in\mathcal{C}}$ a cartesian pseudonatural object of $\mathbb{V}$. This structure is uniquely determined by the universal cowedges.

Proof. For an identity arrow, both the unit comparison and b_{1_U} are morphisms from the initial cowedge to its unit restriction; they agree by uniqueness. For composable arrows $U'' \xrightarrow{v} U' \xrightarrow{u} U$, the composite of b_v with v^*b_u, followed by the associativity constraint, and the map b_{uv} are morphisms from the same initial cowedge over U'' to the same twice-restricted cowedge. They again agree by uniqueness. If all b_u are invertible, reversing them gives the cartesian transition maps; the preceding identities are precisely their pseudonaturality axioms. Any other compatible transitions would invert morphisms with the same universal property and hence coincide with them. □

Remark 4.4. Existence and stability are logically separate. The definition gives b_u only after relative coends exist on both slices; it does not force that map to be invertible. A useful existence theorem must therefore either prove stability or state it as an additional hypothesis.

5 First comparison theorems

Proposition 5.1 (Indexed-equivalence invariance). *Let $F : \mathbb{D} \simeq \mathbb{D}'$ be a chosen pseudonatural adjoint equivalence of indexed categories, including coherent unit and counit. Let $H' : (\mathbb{D}')^\vee \times_{\mathcal{C}} \mathbb{D}' \to \mathbb{V}$, and suppose an indexed natural isomorphism*

$$\theta : H \overset{\cong}{\Longrightarrow} H'\,(F^\vee \times F)$$

has been fixed. Then for every U there is an equivalence

$$\mathbf{Cow}_U(H) \simeq \mathbf{Cow}_U(H')$$

that is the identity on apex objects. In particular, one side has a relative coend if and only if the other does, and the resulting apex objects are canonically isomorphic relative to the chosen equivalence data.

Proof. Transport a component of an H'-cowedge along F_V and precompose with $\theta_{V,A,A}$. Naturality of θ gives fibrewise dinaturality, and its indexed coherence gives base coherence. Applying a chosen pseudoinverse of F gives a functor in the opposite direction. The unit and counit transport to natural isomorphisms between the composites and the identity; their triangle identities give an equivalence of cowedge categories. Because apexes and apex morphisms are unchanged, initial objects correspond and their apexes are uniquely isomorphic by proposition 3.3. □

The word *chosen* in proposition 5.1 is deliberate. An unstructured statement that the fibres merely happen to be equivalent does not provide the coherence needed to transport equation (4).

Proposition 5.2 (Terminal-site reduction). *Let $\mathcal{C} = \mathbf{1}$ with its unique topology. If $\mathbb{D}(*) = \mathcal{A}$, $\mathbb{V}(*) = \mathcal{B}$, and $H_* : \mathcal{A}^{\mathrm{op}} \times \mathcal{A} \to \mathcal{B}$, then a relative cowedge over $*$ is exactly an ordinary cowedge for H_*. Consequently,*

$$\int_{\mathrm{rel},*}^{A\in\mathbb{D}} H(A,A) \cong \int^{A\in\mathcal{A}} H_*(A,A)$$

whenever either side exists.

Proof. The slice $\mathbf{1}/*$ has one object and one arrow. Thus equation (2) is an ordinary family indexed by $A \in \mathcal{A}$, equation (3) is ordinary dinaturality, and equation (4) reduces to the unit axiom. The two cowedge categories are therefore isomorphic. □

Example 5.3 (Fibrewise coends do not automatically assemble). Let $\mathcal{C}$ be the category $0 \xrightarrow{u} 1$, equipped with the trivial topology, and let $\mathbb{V}$ be the constant indexed category **Set**. Define

$$\mathbb{D}(1) = \varnothing, \qquad \mathbb{D}(0) = \mathbf{1},$$

with the unique reindexing functor $u^* : \varnothing \to \mathbf{1}$. Let H_1 be the unique functor with empty domain and let $H_0(*,*) = \{*\}$.

The ordinary fibrewise coend is $\varnothing$ over 1 and a singleton over 0. Since u^* on **Set** is the identity, these two objects cannot be the values of a cartesian indexed object: there is not even a map $\{*\} \to \varnothing$ in the direction of equation (6).

The localized relative coends behave differently. Over 0, a relative cowedge to X is one map $\{*\} \to X$, hence a pointed set. Over 1, the fibre $\mathbb{D}(1)$ contributes no component, but the generalized object $* \in \mathbb{D}(0)$ above $u : 0 \to 1$ again contributes one point of X. Thus $\mathbf{Cow}_0(H)$ and $\mathbf{Cow}_1(H)$ are both the category of pointed sets. Their initial objects are singletons, and the base-change map is the identity. In this example the localization-sensitive definition restores stable assembly precisely by remembering the object visible after pullback.

6 Interface with relative topos theory and prior calculus

6.1 The canonical stack as coefficients

Taking $\mathbb{V} = \mathbb{S}_{(\mathcal{C},J)}$ places the apex of the relative coend over U in the slice topos $\mathcal{E}/\ell_J(U)$. Pullback supplies the reindexing used in equation (4), so stable relative coends form cartesian objects of the canonical stack. This is the direct phase-1 link to relative topos theory. A further identification of such objects with a construction intrinsic to an arbitrary geometric morphism $\mathcal{F} \to \mathcal{E}$ requires a descent theorem and is not used here.

Drieux's indexed hom construction is the motivating special case. In the notation of the slides, its value above U has generalized components

$$[v : V \to U] \longmapsto \mathbb{D}(V)(v^*Y, v^*X), \tag{7}$$

followed, when required, by sheafification. The 2026 slides then formulate an indexed colimit through universal properties on every localization and state an indexed co-Yoneda computation [Dri26a, Dri26b]. Formula equation (7) supports the shape of definition 3.1; the slides do not supply a published proof of a general relative-coend theorem, so none is inferred from them.

6.2 Relation to Shulman's indexed profunctors

For an S-indexed symmetric monoidal category with indexed coproducts, Shulman defines a canceling tensor product. Under Beck–Chevalley, tensor preservation, and fibrewise coequalizer hypotheses, the composite of indexed profunctors is computed by a coend-shaped coequalizer [Shu13, Definition 2.4, Definition 2.10, and Lemma 3.25, pp. 7–8 and 21–22]. This is an established indexed coend calculus and must be treated as the principal comparison theory, not as an anticipated application of the present paper.

The two constructions have different inputs. Shulman's formula begins with an indexed monoidal base and packages fibrewise coproducts together with change-of-base adjoints. Definition 3.1 instead begins with an indexed functor H and enlarges each cowedge over U to every object of $\mathcal{C}/U$. A substantive comparison theorem should identify hypotheses under which Shulman's canceling tensor represents the localized cowedge functor. Until that theorem is proved, the two constructions are related models rather than interchangeable notation.

Betti–Walters is an equally important novelty boundary. Their calculus is formulated for categories enriched in $\mathrm{Span}(\mathcal{E})$, with interchange supplied by the compact closed structure of the span bicategory [BW89]. The construction here is instead written on a site presentation and builds generalized objects in every slice into the cowedge. No equivalence between these formalisms is asserted without a comparison theorem.

6.3 Presentation boundary

An internal category in $\mathcal{E}$, a stack on $(\mathcal{C}, J)$, and a pseudofunctor on a chosen site are connected by comparison results, but they are not definitionally identical. Likewise, equivalent sites can present the same topos without giving literally equal indexing categories. The definition in definition 3.1 is therefore a fixed-presentation construction. Its invariance under indexed equivalence (proposition 5.1) is a first step, but it is not yet a Morita or change-of-site invariance theorem.

7 Roadmap after the foundations

The next two sections settle two of the proof obligations posed by the foundational definition. Theorem 8.2 constructs the relative coend over a fixed base object. Theorem 9.3 proves base-change stability under a pullback and Beck–Chevalley package, while example 9.2 shows that the corresponding claim on an arbitrary site is false. Theorem 10.5 proves Fubini under stable-inner existence. The remaining programme is therefore the following.

Problem 7.1 (Comparison with indexed canceling products)**.** When Shulman's hypotheses hold, determine whether the localized relative coend is represented by his canceling tensor product. State separately the comparison on a fixed site and the additional descent needed for a stack of coefficients.

Problem 7.2 (Relative co-Yoneda)**.** For the indexed hom object suggested by equation (7), identify the coefficient system and tensor needed to type the expression

$$\int_{\mathrm{rel}}^{A} \mathbb{D}(A, X) \otimes F(A),$$

then prove a stable equivalence with $F(X)$. The proof must distinguish presheaf-level computation, sheafification, and descent.

Problem 7.3 (Relative profunctors)**.** Use the existence and Fubini theorems to define composition of localization-sensitive indexed profunctors. Prove associativity and units, and compare the result with Shulman's indexed profunctor equipment and Lee's 2-categorical construction [Lee23].

Problem 7.4 (Site-presentation invariance)**.** Let two small or small-generated sites present equivalent base topoi. Find the additional comparison data on $\mathbb{D}, \mathbb{V}, H$ for which their relative-coend objects correspond. A satisfactory theorem should be invariant under the relevant site Morita equivalence, not merely under a fibrewise equivalence chosen on one presentation.

Together, these obligations retain a clear novelty boundary. The present paper supplies fixed-base computation, a stable range, and interchange. The later stages must connect that calculus to indexed hom objects, profunctor composition, descent, and presentation independence.

8 Existence by generators and relations

Drieux's 2026 slides propose computing indexed colimits from cocompleteness, change-of-base adjoints, and Beck–Chevalley [Dri26b, logical slides 17–18]. The slides do not give a published proof of the localized coend defined in definition 3.1. We therefore begin with a fixed base object and derive the representing object directly. The result is deliberately stated with only the colimits actually used.

8.1 A mate calculation

Assume for the moment that every reindexing functor $v^* : \mathbb{V}(U) \to \mathbb{V}(V)$ has a chosen left adjoint $v_! \dashv v^*$, with unit η^v and counit ε^v. For composable arrows

$$W \xrightarrow{w} V \xrightarrow{v} U,$$

both $(vw)_!$ and $v_!w_!$ are left adjoint, up to the constraint $\alpha_{w,v} : w^*v^* \cong (vw)^*$, to the same functor. Uniqueness of left adjoints gives a canonical natural isomorphism

$$c_{v,w} : (vw)_! \xrightarrow{\cong} v_!w_!.$$

For $Z \in \mathbb{V}(V)$, put

$$\beta_{v,w,Z} : (vw)_! w^* Z \xrightarrow{c_{v,w}} v_! w_! w^* Z \xrightarrow{v_! \varepsilon^w_Z} v_! Z. \tag{8}$$

Lemma 8.1 (The base-coherence mate). *Let $h : Z \to v^* X$, and let $\bar{h} : v_! Z \to X$ be its adjunct. Under $(vw)_! \dashv (vw)^*$, the adjunct of*

$$w^* Z \xrightarrow{w^* h} w^* v^* X \xrightarrow{\alpha_{w,v,X}} (vw)^* X$$

is the composite

$$(vw)_! w^* Z \xrightarrow{\beta_{v,w,Z}} v_! Z \xrightarrow{\bar{h}} X.$$

Proof. The adjunct of h is $\bar{h} = \varepsilon^v_X \, v_! h$. Hence the second displayed composite is

$$(vw)_! w^* Z \xrightarrow{c_{v,w}} v_! w_! w^* Z \xrightarrow{v_! \varepsilon^w_Z} v_! Z \xrightarrow{v_! h} v_! v^* X \xrightarrow{\varepsilon^v_X} X.$$

Naturality of ε^w moves $w_! w^* h$ across ε^w_Z. The result is the counit formula for the composite adjunction $v_! w_! \dashv w^* v^*$. Transporting that composite adjunction along $c_{v,w}$ and $\alpha_{w,v}$ gives the counit formula for $(vw)_! \dashv (vw)^*$. Taking its adjunct therefore yields $\alpha_{w,v,X} w^* h$, as asserted. This is precisely the standard calculus of mates; the displayed computation fixes the direction of every constraint used later. $\square$

8.2 The coequalizer presentation

Fix $U \in \mathcal{C}$. Since $\mathcal{C}$ and every $\mathbb{D}(V)$ are small, the following indexing collections are sets. Define the object of generators

$$P_U = coprod_{\substack{v:V\to U \\ A\in\mathrm{Ob}\mathbb{D}(V)}} v_! H_V(A, A), \tag{9}$$

and write $\iota_{v,A} : v_! H_V(A, A) \to P_U$ for its coproduct injections.

There are two sorts of relations. Fibrewise dinaturality is generated by

$$R^{\mathrm{din}}_U = \coprod_{\substack{v:V\to U \\ f:A\to B \text{ in } \mathbb{D}(V)}} v_! H_V(B, A). \tag{10}$$

On the summand indexed by $(v, f : A \to B)$, define the parallel maps to P_U by

$$d^{\mathrm{din}}_0 = \iota_{v,A} \, v_! H_V(f^{\mathrm{op}}, 1_A), \tag{11}$$

$$d^{\mathrm{din}}_1 = \iota_{v,B} \, v_! H_V(1_B, f). \tag{12}$$

Base coherence is generated by

$$R^{\mathrm{bas}}_U = \coprod_{\substack{v:V\to U,\ w:W\to V \\ A\in\mathrm{Ob}\mathbb{D}(V)}} (vw)_! w^* H_V(A, A). \tag{13}$$

On the summand indexed by (v, w, A), its two maps to P_U are

$$d^{\mathrm{bas}}_0 = (vw)_! w^* H_V(A, A) \xrightarrow{(vw)_! \chi^H_{w,A,A}} (vw)_! H_W(w^* A, w^* A) \xrightarrow{\iota_{vw,w^*A}} P_U, \tag{14}$$

$$d^{\mathrm{bas}}_1 = (vw)_! w^* H_V(A, A) \xrightarrow{\beta_{v,w,H_V(A,A)}} v_! H_V(A, A) \xrightarrow{\iota_{v,A}} P_U. \tag{15}$$

Let

$$R_U = R^{\mathrm{din}}_U \amalg R^{\mathrm{bas}}_U, \qquad d_i = d^{\mathrm{din}}_i \amalg d^{\mathrm{bas}}_i quad (i = 0, 1).$$

Theorem 8.2 (Fixed-base coequalizer formula)**.** *Suppose that every v^* has a chosen left adjoint $v_!$, and that $\mathbb{V}(U)$ has the coproducts in equations (9), (10) and (13) and the coequalizer*

$$R_U \overset{d_0}{\underset{d_1}{\rightrightarrows}} P_U \xrightarrow{q_U} L_U. \tag{16}$$

Then L_U, with the cowedge defined below, is the relative coend of H over U. In particular,

$$\mathbb{V}(U)(L_U, X) \cong \mathrm{Cow}^{\mathrm{rel}}_U(H; X)$$

naturally in $X \in \mathbb{V}(U)$.

No pullbacks in $\mathcal{C}$, Beck–Chevalley isomorphisms, or preservation of colimits by reindexing are needed for this fixed-base statement.

Proof. We give the construction and both directions of the representing bijection.

Step 1: the universal components. For each $v : V \to U$ and $A \in \mathbb{D}(V)$, transpose

$$v_! H_V(A, A) \xrightarrow{\iota_{v,A}} P_U \xrightarrow{q_U} L_U$$

through $v_! \dashv v^*$. Denote the resulting map by

$$\lambda^U_{v,A} : H_V(A, A) \longrightarrow v^* L_U. \tag{17}$$

Step 2: fibrewise dinaturality. For $f : A \to B$ in $\mathbb{D}(V)$, the equation $q_U d_0^{\mathrm{din}} = q_U d_1^{\mathrm{din}}$ on the (v, f)-summand says

$$q_U \iota_{v,A} v_! H_V(f^{\mathrm{op}}, 1_A) = q_U \iota_{v,B} v_! H_V(1_B, f).$$

Taking adjuncts under $v_! \dashv v^*$ gives exactly

$$\lambda^U_{v,A} H_V(f^{\mathrm{op}}, 1_A) = \lambda^U_{v,B} H_V(1_B, f).$$

Step 3: base coherence. On the (v, w, A)-summand of R^{bas}_U, the equation $q_U d_0^{\mathrm{bas}} = q_U d_1^{\mathrm{bas}}$ has two adjuncts under $(vw)_! \dashv (vw)^*$. The adjunct of the first is

$$w^* H_V(A, A) \xrightarrow{\chi^H_{w,A,A}} H_W(w^*A, w^*A) \xrightarrow{\lambda^U_{vw,w^*A}} (vw)^* L_U.$$

By lemma 8.1, the adjunct of the second is

$$w^* H_V(A, A) \xrightarrow{w^* \lambda^U_{v,A}} w^* v^* L_U \xrightarrow{\alpha_{w,v,L_U}} (vw)^* L_U.$$

Their equality is precisely equation (4). Thus λ^U is a relative cowedge.

Step 4: maps out of the coequalizer give cowedges. Given $m : L_U \to X$, define

$$\omega^m_{v,A} = v^* m \, \lambda^U_{v,A}.$$

The equations already proved for λ^U, together with naturality of the pseudofunctor constraints, show that ω^m is a relative cowedge.

Step 5: every cowedge factors uniquely. Conversely, let $\omega \in \mathrm{Cow}^{\mathrm{rel}}_U(H; X)$. Transpose each component to

$$\bar{\omega}_{v,A} : v_! H_V(A, A) \longrightarrow X.$$

The coproduct property gives a unique $\bar{\omega} : P_U \to X$ with $\bar{\omega} \iota_{v,A} = \bar{\omega}_{v,A}$. On a summand of R^{din}_U, the equality $\bar{\omega} d_0 = \bar{\omega} d_1$ is, after adjunction, exactly the fibrewise dinaturality of ω. On a summand of R_U^{bas}, lemma 8.1 identifies the same equality with the base-coherence equation for ω. Thus $\bar{\omega}$ coequalizes d_0, d_1, so there is a unique $m : L_U \to X$ such that $mq_U = \bar{\omega}$. Taking adjuncts on each generator shows $v^* m\, \lambda^U_{v,A} = \omega_{v,A}$.

The two assignments are inverse because both are determined on the coproduct generators, and they are natural in X because postcomposition is used at every step. Proposition 3.3 now identifies L_U as the relative coend. □

Remark 8.3 (Relation to the ordinary formula). When $\mathcal{C} = \mathbf{1}$, the base-relation summands only encode unit constraints, and equation (16) reduces to the standard coequalizer by objects and arrows of $\mathbb{D}(*)$. This agrees with the ordinary definition as an initial cowedge [Lor23, Definitions 1.1.4 and 1.1.6, pp. 23–24]. On a nonterminal base, the summand R_U^{bas} is additional data; deleting it changes the universal problem.

9 Base-change stability and the missing hypothesis

The fixed-base construction does not by itself imply that the maps $b_u : L_{U'} \to u^* L_U$ are invertible. The exact criterion is immediate from the universal property.

Proposition 9.1 (Exact stability criterion)**.** *For $u : U' \to U$, assume that relative coends exist over U and U'. The canonical map b_u is an isomorphism if and only if the restricted universal cowedge*

$$\rho_u(L_U, \lambda^U) = (u^* L_U, \rho_u \lambda^U)$$

is initial in $\mathbf{Cow}_{U'}(H)$.

Proof. If b_u is invertible, it is an isomorphism in $\mathbf{Cow}_{U'}(H)$ from the initial cowedge $(L_{U'}, \lambda^{U'})$ to the restricted cowedge; hence the latter is initial. Conversely, if both cowedges are initial, there is a unique arrow in each direction between them. The two composites are endomorphisms of initial objects and therefore identities. The arrow from $(L_{U'}, \lambda^{U'})$ is, by its defining equation, b_u; hence it is invertible. □

The following example isolates a genuine gap in a tempting statement of the hypotheses.

Example 9.2 (Available Beck–Chevalley squares are not enough)**.** Let $\mathcal{C}$ have three objects and two nonidentity arrows

$$0 \xrightarrow{p} 2 \xleftarrow{q} 1,$$

and equip it with the trivial Grothendieck topology. Let $\mathbb{V}$ be the constant indexed category **Set**, so every reindexing functor and every chosen left adjoint is the identity. All coproducts and coequalizers exist and are preserved, and the Beck–Chevalley map for every pullback square that exists is an identity.

Put

$$\mathbb{D}(2) = \varnothing, \qquad \mathbb{D}(0) = \mathbb{D}(1) = \mathbf{1},$$

with the unique reindexing functors, and set $H_0(*,*) = H_1(*,*) = \mathbf{1}$. There is no component over 2.

A relative cowedge over 2 with apex X consists of one point of X contributed by $(p,*)$ and one independently contributed by $(q,*)$. There is no arrow in a slice relating these two components. Consequently

$$\mathrm{Cow}^{\mathrm{rel}}_2(H; X) \cong X \times X \quad \text{and} \quad L_2 \cong \{0, 1\}.$$

Over 0 and 1 there is one component, so $L_0 \cong L_1 \cong \mathbf{1}$. Since reindexing in $\mathbb{V}$ is the identity,

$$b_p : \mathbf{1} \longrightarrow \{0,1\}, \qquad b_q : \mathbf{1} \longrightarrow \{0,1\}$$

are the two coproduct injections and are not isomorphisms.

Thus cocomplete fibres, cocontinuous reindexing, left adjoints, and Beck–Chevalley for all *available* pullback squares do not imply stability on an arbitrary site. What is missing here is the pullback of p and q. The next theorem shows that chosen pullbacks, together with the expected exactness, close precisely this gap for the displayed presentation.

Theorem 9.3 (Stable base change from pullbacks)**.** *Assume the hypotheses of theorem 8.2 for every base object. Assume in addition that:*

(S1) $\mathcal{C}$ has chosen pullbacks;

(S2) the chosen adjoints satisfy Beck–Chevalley: for every chosen pullback square

$$\begin{tikzcd} W \arrow[r, "r"] \arrow[d, "s"'] & V \arrow[d, "v"] \\ U' \arrow[r, "u"'] & U \end{tikzcd} \qquad \mathrm{BC}_{u,v} : s_! r^* \xrightarrow{\cong} u^* v_!; \tag{18}$$

(S3) every u^* preserves the coproducts and coequalizers appearing in equations (9), (10), (13) and (16).

Then every canonical base-change map

$$b_u : L_{U'} \xrightarrow{\cong} u^* L_U$$

is an isomorphism. Hence the fixed-base coequalizers assemble, uniquely, to a cartesian pseudonatural object of $\mathbb{V}$.

Proof. Fix $u : U' \to U$. We construct an inverse $a_u : u^* L_U \to L_{U'}$ on generators and then check both composites.

Step 1: pull a generator across the square. For every $v : V \to U$, choose its pullback along u as in equation (18). Preservation of the coproduct in equation (9), Beck–Chevalley, and the indexed comparison of H give

$$u^* P_U \cong \coprod_{v:V\to U,\ A\in\mathbb{D}(V)} u^* v_! H_V(A,A) \tag{19}$$

$$\xrightarrow{\coprod \mathrm{BC}_{u,v}^{-1}} \coprod_{v,A} s_! r^* H_V(A,A) \tag{20}$$

$$\xrightarrow{\coprod s_! \chi^H_{r,A,A}} \coprod_{v,A} s_! H_W(r^*A, r^*A) \xrightarrow{\coprod \iota_{s,r^*A}} P_{U'} \xrightarrow{q_{U'}} L_{U'}. \tag{21}$$

Call this composite $a_P : u^* P_U \to L_{U'}$.

Step 2: the relations are respected. For a dinaturality summand indexed by $(v, f : A \to B)$, pullback turns f into $r^* f : r^* A \to r^* B$. Under the first three isomorphisms in equation (21), the two maps are exactly the two legs of the dinaturality relation indexed by $(s, r^* f)$ in $R^{\mathrm{din}}_{U'}$. They therefore agree after $q_{U'}$.

For a base-relation summand indexed by $v : V \to U$, $w : T \to V$, and $A \in \mathbb{D}(V)$, form the chosen pullbacks

$$W = U' \times_U V, \qquad Z = U' \times_U T.$$

Writing $(s, r) : W \to U' \times V$ and $(t, k) : Z \to U' \times T$, the pullback universal property gives a unique $\bar{w} : Z \to W$ with

$$s\bar{w} = t, \qquad r\bar{w} = wk.$$

Pasting of Beck–Chevalley squares identifies the pullback of the first leg equation (14) with the generator $(t, \bar{w}^*r^*A)$, and identifies the pullback of the second leg equation (15) with the generator (s, r^*A). The comparison $\bar{w}^*r^*A \cong k^*w^*A$ is the pseudofunctorial constraint of $\mathbb{D}$; the analogous comparison for H is the coherence of χ^H. More explicitly, after these identifications the two composites into $L_{U'}$ are

$$t_!\bar{w}^*H_W(r^*A, r^*A) \xrightarrow{t_!\chi^H_{\bar{w},r^*A,r^*A}} t_!H_Z(\bar{w}^*r^*A, \bar{w}^*r^*A) \xrightarrow{q_{U'}\iota_{t,\bar{w}^*r^*A}} L_{U'},$$

$$t_!\bar{w}^*H_W(r^*A, r^*A) \xrightarrow{\beta_{s,\bar{w},H_W(r^*A,r^*A)}} s_!H_W(r^*A, r^*A) \xrightarrow{q_{U'}\iota_{s,r^*A}} L_{U'}.$$

They are precisely the two legs of the base relation in $R^{\mathrm{bas}}_{U'}$ indexed by $(s, \bar{w}, r^*A)$. The second leg has the displayed direction by lemma 8.1. Thus it too is equalized by $q_{U'}$.

It follows that a_P coequalizes u^*d_0, u^*d_1. Since u^* preserves equation (16), there is a unique

$$a_u : u^*L_U \longrightarrow L_{U'} \quad \text{with} \quad a_u, u^*q_U = a_P. \tag{22}$$

Step 3: $a_ub_u = 1$. It suffices to precompose with every coproduct generator $v'_!H_{V'}(A', A') \to P_{U'} \to L_{U'}$, since a coequalizer is an epimorphism. Pull back the composite $uv' : V' \to U$ along u:

$$W = U' \times_U V' \xrightarrow{r} V', \qquad W \xrightarrow{s} U'.$$

There is a canonical section $\delta = (v', 1_{V'}) : V' \to W$ with $s\delta = v'$ and $r\delta = 1_{V'}$. By the definition of b_u, the image of the (v', A')-generator is the mate of the universal component $\lambda^U_{uv',A'}$. Applying a_u converts it, by Beck–Chevalley, into the (s, r^*A')-generator. The mate along the section δ is exactly the base relation in $L_{U'}$ indexed by (s, δ, r^*A'); it identifies that generator with $(v', \delta^*r^*A') \cong (v', A')$. Hence a_ub_u is the identity on all generators and therefore on $L_{U'}$.

Step 4: $b_ua_u = 1$. Because u^* preserves the coequalizer, u^*q_U is an epimorphism. Using the coproduct and Beck–Chevalley decomposition in equation (21), it is enough to check the summand associated to a pullback square equation (18) and $A \in \mathbb{D}(V)$. The map a_u sends it to the (s, r^*A)-generator of $L_{U'}$. The map b_u then sends that generator to the restriction of λ^U_{us,r^*A}. Since $us = vr$, the base relation of L_U indexed by (v, r, A) identifies this component with the original (v, A)-generator. Pulling that equality back along u proves that b_ua_u is the identity on the chosen summand. Hence it is the identity on u^*L_U.

Thus $a_u = b_u^{-1}$. Coherent assembly now follows from theorem 4.3. □

Remark 9.4 (Exact comparison with Shulman's hypotheses). Shulman calls the existence of $f_! \dashv f^*$ together with Beck–Chevalley for pullback squares "indexed coproducts" [Shu13, Definition 2.4, p. 7]. His tensor-preservation condition is Definition 2.10, and indexed profunctor composition is the coequalizer of Lemma 3.25 [Shu13, pp. 8 and 21–22]. Theorem 9.3 is not a restatement of that lemma: our R^{bas}_U explicitly imposes relations between all generalized objects in $\mathcal{C}/U$, while Shulman's formula composes enriched indexed profunctors. A comparison between the resulting objects still requires a typed identification of these two universal problems.

10 A Fubini theorem for stable relative coends

The ordinary Fubini theorem identifies a coend over a product category with either order of iterated coends [Lor23, Theorem 1.3.1, pp. 37–39]. In the relative setting an inner coend must itself vary as indexed data before an outer relative coend can even be typed. This is why stability, rather than mere fibrewise existence, is the correct hypothesis.

Let

$$\mathbb{A}, \mathbb{B} : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Cat}$$

be small-fibred indexed categories, and let

$$H : \mathbb{A}^{\vee} \times_{\mathcal{C}} \mathbb{A} \times_{\mathcal{C}} \mathbb{B}^{\vee} \times_{\mathcal{C}} \mathbb{B} \longrightarrow \mathbb{V} \tag{23}$$

be an indexed functor. Put

$$\widehat{H}_V\big((A', B'), (A, B)\big) = H_V(A', A, B', B).$$

Then $\widehat{H}$ is an indexed functor on $(\mathbb{A} \times_{\mathcal{C}} \mathbb{B})^{\vee} \times_{\mathcal{C}} (\mathbb{A} \times_{\mathcal{C}} \mathbb{B})$.

Definition 10.1 (Simultaneous relative cowedge)**.** For $U \in \mathcal{C}$ and $X \in \mathbb{V}(U)$, a simultaneous relative cowedge for H is a family

$$\omega_{v,A,B} : H_V(A, A, B, B) \longrightarrow v^*X$$

for $v : V \to U$, $A \in \mathbb{A}(V)$, and $B \in \mathbb{B}(V)$, which is dinatural separately in A and in B, and satisfies the joint base-coherence equation

$$\omega_{vw,w^*A,w^*B}\, \chi^{H}_{w,A,A,B,B} = \alpha_{w,v,X}\, w^*\omega_{v,A,B}.$$

Its set is denoted $\mathrm{Cow}^{\mathrm{rel,bi}}_U(H; X)$.

Lemma 10.2 (Product versus separate dinaturality)**.** *There is a natural equality of cowedge sets*

$$\mathrm{Cow}^{\mathrm{rel}}_U(\widehat{H}; X) = \mathrm{Cow}^{\mathrm{rel,bi}}_U(H; X).$$

Proof. Dinaturality for a morphism $(f, g) : (A, B) \to (A', B')$ in a product fibre implies the two separate equations by taking g or f to be an identity. Conversely, factor

$$(f, g) = (1_{A'}, g)(f, 1_B).$$

Apply A-dinaturality to the first change of object and B-dinaturality to the second; functoriality of H_V makes the middle terms agree. This gives dinaturality for (f, g). The base-coherence equation is literally the same for $\widehat{H}$, so the two sets coincide, naturally in X. □

For $B', B \in \mathbb{B}(U)$, restrict H to the slice $\mathcal{C}/U$ by setting

$$H^{B',B}_w(A', A) = H_W(A', A, w^*B', w^*B) \qquad (w : W \to U).$$

Assume that its relative coend exists and write

$$K_U(B', B) = \int_{\mathrm{rel},U}^{A\in\mathbb{A}} H(A, A, B', B). \tag{24}$$

Its universal components are

$$\kappa^{U,B',B}_{w,A} : H_W(A, A, w^*B', w^*B) \longrightarrow w^*K_U(B', B). \tag{25}$$

Lemma 10.3 (Stable inner coends form an indexed bifunctor)**.** *Suppose the objects in equation (24) exist for every U, B', B, and their canonical base-change maps*

$$b_u^{B',B} : K_{U'}(u^*B', u^*B) \longrightarrow u^*K_U(B', B) \tag{26}$$

are isomorphisms. Then the K_U assemble to an indexed functor

$$K : \mathbb{B}^\vee \times_{\mathcal{C}} \mathbb{B} \longrightarrow \mathbb{V},$$

whose comparison along u is $(b_u^{B',B})^{-1}$.

Proof. First fix U. For arrows $r : B'_0 \to B'_1$ and $s : B_0 \to B_1$ in $\mathbb{B}(U)$, define

$$K_U(r^{\mathrm{op}}, s) : K_U(B'_1, B_0) \longrightarrow K_U(B'_0, B_1)$$

as the unique map induced by the inner cowedge whose (w, A)-component is

$$\begin{aligned} H_W(A, A, w^*B'_1, w^*B_0) &\xrightarrow{H_W(1,1,(w^*r)^{\mathrm{op}},w^*s)} H_W(A, A, w^*B'_0, w^*B_1) \\ &\xrightarrow{\kappa_{w,A}^{U,B'_0,B_1}} w^*K_U(B'_0, B_1). \end{aligned}$$

Naturality of H shows that this is an inner relative cowedge. Two successive such maps and the map associated to the composite have identical composites with every universal component $\kappa_{w,A}$, so uniqueness in the inner universal property makes them equal. The same argument with identity arrows proves the identity law. Thus each K_U is a bifunctor with the asserted variance.

For $u : U' \to U$, set

$$\chi^K_{u,B',B} = (b_u^{B',B})^{-1} : u^*K_U(B', B) \xrightarrow{\cong} K_{U'}(u^*B', u^*B).$$

To check naturality in B', B, compare the two candidate maps after precomposition with every component κ. Both reduce to the indexed naturality square of H, followed by the same universal component; hence they agree by uniqueness. The unit and composition constraints for χ^K are the inverses of the canonical base-change constraints. Their coherence is theorem 4.3, or equivalently the same uniqueness argument applied to a twice-restricted universal cowedge. Therefore K is an indexed functor. ☐

Proposition 10.4 (Currying simultaneous cowedges)**.** *Under the hypotheses of lemma 10.3, there is a natural bijection*

$$\mathrm{Cow}_U^{\mathrm{rel,bi}}(H; X) \cong \mathrm{Cow}_U^{\mathrm{rel}}(K; X) \tag{27}$$

for every U and $X \in \mathbb{V}(U)$.

Proof. We construct both maps and verify every axiom.

From a simultaneous cowedge to an outer cowedge. Let ω be simultaneous. Fix $v : V \to U$ and $B \in \mathbb{B}(V)$. For $w : W \to V$ and $A \in \mathbb{A}(W)$, define

$$H_W(A, A, w^*B, w^*B) \xrightarrow{\omega_{vw,A,w^*B}} (vw)^*X \xrightarrow{\alpha^{-1}_{w,v,X}} w^*v^*X. \tag{28}$$

The A-dinaturality of ω proves fibrewise dinaturality of this family. For $z : Z \to W$, its base-coherence square is the joint base-coherence square of ω, pasted with the associativity pentagon for α. Hence equation (28) is an inner relative cowedge over V with apex v^*X. Universality of $K_V(B, B)$ supplies a unique map

$$\bar{\omega}_{v,B} : K_V(B, B) \longrightarrow v^*X \tag{29}$$

whose pullback along every w, after $\kappa^{V,B,B}_{w,A}$, is equation (28).

We check B-dinaturality. For $f : B \to C$ in $\mathbb{B}(V)$, the two required maps from $K_V(C, B)$ to v^*X are

$$\bar{\omega}_{v,B} K_V(f^{\mathrm{op}}, 1_B), \qquad \bar{\omega}_{v,C} K_V(1_C, f).$$

Precompose their pullbacks with every universal component of $K_V(C, B)$. By the definition of the two actions of K, the resulting maps are respectively

$$H_W(A, A, w^*C, w^*B) \xrightarrow{H(1,1,(w^*f)^{\mathrm{op}},1)} H_W(A, A, w^*B, w^*B) \longrightarrow w^*v^*X,$$
$$H_W(A, A, w^*C, w^*B) \xrightarrow{H(1,1,1,w^*f)} H_W(A, A, w^*C, w^*C) \longrightarrow w^*v^*X.$$

They agree by the B-dinaturality of ω. The representing property of the inner coend then makes the two maps from $K_V(C, B)$ equal.

For base coherence, let $w : W \to V$. We must show

$$\bar{\omega}_{vw,w^*B}\, \chi^K_{w,B,B} = \alpha_{w,v,X}\, w^*\bar{\omega}_{v,B} : w^*K_V(B, B) \longrightarrow (vw)^*X. \tag{30}$$

Precompose both sides with $b^{B,B}_w : K_W(w^*B, w^*B) \to w^*K_V(B, B)$, which is an isomorphism. Then precompose with every $\kappa^{W,w^*B,w^*B}_{z,A}$. The defining equation for the canonical base-change map $b^{B,B}_w$ converts the right-hand side into the restriction of equation (28); the left-hand side is the same map because $\chi^K_{w,B,B} b^{B,B}_w = 1$. After cancelling the associativity constraints, equality is exactly the joint base coherence of ω. Hence equation (30) holds, and $\bar{\omega}$ is an outer relative cowedge for K.

From an outer cowedge to a simultaneous cowedge. Conversely, let $\tau \in \mathrm{Cow}^{\mathrm{rel}}_U(K; X)$. Define

$$\omega^\tau_{v,A,B} : H_V(A, A, B, B) \xrightarrow{\kappa^{V,B,B}_{1_V,A}} K_V(B, B) \xrightarrow{\tau_{v,B}} v^*X, \tag{31}$$

with the unit constraints suppressed. The inner cowedge equation for κ gives A-dinaturality. The definition of the two actions of K, followed by the B-dinaturality of τ, gives B-dinaturality. For $w : W \to V$, the compatibility of κ with the stable comparison χ^K_w, followed by the outer base-coherence equation for τ, gives

$$\omega^\tau_{vw,w^*A,w^*B} \chi^H_{w,A,A,B,B} = \alpha_{w,v,X} w^*\omega^\tau_{v,A,B}.$$

Thus ω^τ is simultaneous.

The constructions are inverse. Starting with ω, the uncurried component at (v, A, B) evaluates the map $\bar{\omega}_{v,B}$ on the identity-base universal component. Its defining property equation (28) therefore returns $\omega_{v,A,B}$, up to the unit constraint. Starting with τ, the inner cowedge used to construct $\bar{\omega}_{v,B}$ from ω^τ is the restriction of $\tau_{v,B}$ along all $w : W \to V$; this is the outer base-coherence axiom for τ. The uniqueness clause for the inner coend therefore gives $\bar{\omega}_{v,B} = \tau_{v,B}$. Both assignments commute with postcomposition in X, proving naturality. □

Theorem 10.5 (Relative Fubini)**.** *Assume that the $\mathbb{A}$-inner coends $K_U(B', B)$ of equation (24) exist and are stable under base change. Then the following two objects exist simultaneously, and when they exist there is a canonical isomorphism*

$$\int^{(A,B)\in\mathbb{A}\times\mathbb{B}}_{\mathrm{rel},U} H(A, A, B, B) \;\cong\; \int^{B\in\mathbb{B}}_{\mathrm{rel},U} \left(\int^{A\in\mathbb{A}}_{\mathrm{rel}} H(A, A, B, B) \right). \tag{32}$$

Here the right-hand side means the relative coend of the indexed bifunctor K, not a list of unrelated fibrewise coends.

If, symmetrically, the $\mathbb{B}$-inner coends exist stably, then whenever one of the simultaneous or iterated coends exists, all corresponding admissible orders exist and are canonically isomorphic. For three or more variables, all comparison composites between admissible iterated orders agree.

Proof. By lemma 10.2 and proposition 10.4, for every $X \in \mathbb{V}(U)$ there are natural bijections

$$\mathrm{Cow}_U^{\mathrm{rel}}(\widehat{H}; X) = \mathrm{Cow}_U^{\mathrm{rel,bi}}(H; X) \cong \mathrm{Cow}_U^{\mathrm{rel}}(K; X).$$

The simultaneous coend is, by proposition 3.3, a representing object for the first functor of X; the iterated coend is a representing object for the last. The natural bijection therefore transports a universal element in either direction. Hence one object exists if and only if the other does, and uniqueness of representing objects gives the canonical isomorphism equation (32).

Repeating the proof with $\mathbb{A}$ and $\mathbb{B}$ interchanged gives the other iterated order. Every comparison is the unique isomorphism that transports the same simultaneous universal cowedge. For three or more variables, any two composites of adjacent Fubini comparisons have that same property and hence are equal by uniqueness. This proves all associativity and higher interchange coherences without an additional diagram chase. □

Remark 10.6 (Why stability is not cosmetic). If the inner objects $K_U(B', B)$ merely exist separately, the canonical maps in equation (26) need not be invertible. Then they do not provide the indexed comparison in the direction required for an indexed functor K, so the outer relative coend in equation (32) is not typed by the present definition. The theorem therefore does not silently promote fibrewise existence to indexed Fubini. Under the hypotheses of theorem 9.3, the needed stability follows automatically from the coequalizer presentation.

11 Boundary of phases 2–3

Theorems 8.2, 9.3, and 10.5 establish fixed-base existence, one checkable stable range, and Fubini. They do not yet identify the relative coend with Shulman's canceling tensor, prove the relative co-Yoneda formula announced in Drieux's slides, construct a profunctor equipment, or remove dependence on a chosen site presentation. Those claims remain exactly the problems listed in section 7.

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